Technical University of Košice
Faculty of Electrical Engineering and Informatics

Implementation of augmentation methods for medical ultrasound data

Bachelor's Thesis

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Study Programme:	Intelligent Systems
Field of Study:	Computer Science
Department:	Department of Cybernetics and Artificial Intelligence (DCAI)
Supervisor:	doc. Ing. Marek Bundzel, PhD.
Consultant:	Ing. Maros Hliboky

Košice 2024 **Volodymyr Novokhatskyi**

Abstract in English

Medical imaging, particularly ultrasound, is crucial for diagnosing diseases. However, the limited availability of annotated medical image data challenges the training of effective deep learning models. This thesis explores data augmentation techniques to enhance deep learning model performance in classifying lung ultrasound images. Various single and combined augmentation methods were evaluated, including affine and spatial transformations, pixel-level transformations, and random erasing. A mechanism for automatic and parallel verification of proposed experiments was created. The results showed significant improvement with data augmentation. The best-performing single method, a 30-degree rotation, achieved an F1-mean score of 0.365, compared to 0.149 to 0.249 without augmentation. Combined strategies further enhanced performance, with the best combination yielding an F1-mean score of 0.486. Mirrored combinations outperformed their original counterparts, highlighting the importance of method order. This thesis concludes that strategic application and combination of data augmentation techniques significantly improve the robustness and accuracy of deep learning models for medical ultrasound analysis, supporting better diagnostic outcomes and patient care.

Keywords in English

medical image processing, lung ultrasound, data augmentation, deep learning

Abstract in Slovak

Medicínske zobrazovanie, najmä ultrazvuk, je kľúčové pre diagnostiku chorôb. Nedostatok anotovaných dát z medicínskych snímok však predstavuje výzvu pre trénovanie efektívnych modelov hlbokého učenia. Táto diplomová práca skúma techniky augmentácie dát s cieľom zlepšiť výkon modelov hlbokého učenia pri klasifikácii pľúcnych ultrazvukových snímok. Boli vyhodnotené rôzne samostatné a kombinované metódy augmentácie, vrátane afinných a priestorových transformácií, pixelových transformácií a náhodného vymazávania. Bol vytvorený mechanizmus na automatické a paralelné overovanie navrhovaných experimentov. Výsledky ukázali významné zlepšenie pomocou augmentácie dát. Najlepšie výkonná samostatná metóda, rotácia o 30 stupňov, dosiahla skóre F1-mean 0,365, v porovnaní s 0,149 až 0,249 bez augmentácie. Kombinované stratégie ďalej zlepšili výkon, pričom najlepšia kombinácia dosiahla skóre F1-mean 0,486. Kombinácie

so zrkadlením prekonali svoje pôvodné náprotivky, čo zdôrazňuje dôležitosť poradia metód. Táto práca uzatvára, že strategické použitie a kombinácia techník augmentácie dát významne zlepšujú robustnosť a presnosť modelov hlbokého učenia pre analýzu medicínskych ultrazvukových snímok, čo podporuje lepšie diagnostické výsledky a starostlivosť o pacientov.

Keywords in Slovak

spracovanie medicínskych obrazov, ultrazvuk pľúc, augmentácia dát, hlboké učenie

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TECHNICAL UNIVERSITY OF KOŠICE

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BACHELOR THESIS ASSIGNMENT

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Implementation of augmentation methods for medical ultrasound data

Implementácia augmentačných metód pre medicínske ultrazvukové dáta

Student: **Volodymyr Novokhatskyi**

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Thesis preparation instructions:

1. Create a review on the augmentation of image data associated with training neural networks, focusing on medical image data.
2. Provide an overview of existing solutions for parallel tuning of hyperparameters in neural networks.
3. Utilize processed insights to design augmented methods suitable for ultrasound image data.
4. Apply selected methods and create a mechanism for automatic and parallel verification of proposed experiments.
5. Compare the results of individual augmentation techniques and perform appropriate evaluation.
6. Prepare documentation according to the supervisor's instructions.

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prof. Ing. Liberios Vokorokos, PhD.

Dean of the Faculty

Declaration

I hereby declare that this thesis is my own work and effort. Where other sources of information have been used, they have been acknowledged.

Košice, 26.5.2024 . . . .

Signature

Acknowledgement

With deepest gratitude, I acknowledge the invaluable guidance and support of Ing. Maros Hliboky, who served as my consultant throughout this bachelor's thesis. His expertise in the field and unwavering dedication significantly impacted the successful completion of this project.

I extend my sincere thanks to my supervisor, doc. Ing. Marek Bundzel, PhD., for his insightful guidance, unwavering encouragement, and valuable contributions.

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Introduction

Medical imaging, particularly ultrasound, plays a crucial role in healthcare for diagnosing and monitoring diseases. However, the limited annotated medical image data hampers the effective training of deep learning models. Data augmentation offers a solution by artificially expanding the dataset, improving model robustness and generalization. This thesis tackles the challenge of augmenting ultrasound images to enhance deep learning model performance.

While augmentation is successful in natural image processing, its application in medical imaging requires preserving diagnostic features and avoiding artifacts. There is currently a strong focus on studying the impact of data augmentation in the field of medical imaging. However, there are relatively few studies directed specifically at ultrasound imaging, particularly of the lungs, due to its novelty and the lack of precise definition and standardization of the experimented methods.

This thesis aims to address these gaps by implementing and evaluating various data augmentation techniques. It reviews existing methods, addresses key challenges, and explores parallel hyperparameter tuning. Effectiveness is assessed through systematic experiments, aiming to improve classification performance of ultrasound images.

An additional focus of this thesis is on the evaluation of combined data augmentation methods. The study examines the potential of combining the best-performing augmentation techniques to enhance model performance. Furthermore, the order of applying these methods is explored, highlighting its influence on the outcomes. These insights suggest that strategic combinations and orderings of augmentation techniques can lead to substantial improvements in model robustness and diagnostic accuracy.

By overcoming these challenges, this work contributes to developing more accurate and reliable deep learning models for medical image analysis, supporting better diagnostic outcomes and patient care.

1 Task formulation

Our objective in this study is to :

1. Create a review on the augmentation of image data associated with training neural networks, focusing on medical image data (Chapter 2: Analytical Chapter):
   - Introduce the concept of image data augmentation and its necessity in training neural networks.
   - Clarify challenges and considerations when augmenting medical images.
   - Discuss the different types of image data augmentation techniques used in a medical domain.
2. Provide an overview of existing solutions for parallel tuning of hyperparameters in neural networks (Chapter 2: Analytical Chapter):
   - Describe Concept and Importance of Hyperparameter Tuning
   - Outline Methods of Hyperparameter Tuning
   - Disscuss existing Solutions for Parallel Hyperparameter Tuning
3. Utilize processed insights to design augmented methods suitable for ultrasound image data (Chapter 3: Synthetic Chapter):
   - Introduce the dataset used in the study.
   - Discuss the pre-processing steps applied to the dataset and its characteristics
   - Define Specific Augmentation Technique with Parameter Ranges
4. Apply selected methods and create a mechanism for automatic and parallel verification of proposed experiments (Chapter 3: Synthetic Chapter):

- Design and Implementation of an Automatic Execution Mechanism
- Implementation of Parallel Processing Capabilities
- Setting Up and Managing Remote Connections for Distributed Processing

5. Compare the results of individual augmentation techniques and perform appropriate evaluation (Chapter 4: Evaluation):

    - Define Evaluation Criteria
    - Conduct Systematic Experiments to Compare Augmentation Methods
    - Summarize Findings and Identify Most Effective Augmentation Methods

6. Prepare documentation according to the supervisor's instructions (Chapter 5: Conclusion):

    - Provide a summary of the study's findings and discuss their implications for medical data processing.
    - Provide recommendations for further improvement to proposed data augmentation strategies
    - Gather all relevant documentation and deliver it as per the consultant's instructions.

2 Analytical chapter

This chapter delves into the analytical aspects of data augmentation in the context of medical image processing. We will discuss the foundational concepts of data augmentation, the unique challenges posed by medical images.

Furthermore, we will explore methods for parallel hyperparameter tuning and the criteria for selecting appropriate neural network architectures.

2.1 Data Augmentation

Large datasets are essential for training accurate and reliable neural networks, a concept well-established in machine learning [1]. However, collecting a vast number of images comparable to those in datasets like ImageNet is often challenging or impossible due to the specific nature of the images required. This limitation necessitates the exploration of alternative solutions with the available resources.

Data augmentation emerges as a critical strategy in this context, aimed at expanding the size of training datasets by generating modified versions of existing data. Despite the availability of large datasets, the significance of augmentation techniques cannot be overlooked. These methods enable manipulations within the image data space, creating alternative variants that account for potential invariances introduced by variables such as shooting angle and distance [2].

2.1.1 Challenges with Augmenting Medical Images

Augmenting medical images for training deep learning models is crucial for enhancing model accuracy and robustness. However, this process involves specific challenges that need careful consideration to maintain the reliability and usefulness of the models in clinical settings. This section explores these challenges, emphasizing the preservation of label accuracy, the specificity of anatomical features, data integrity, the avoidance of diagnostic-compromising artifacts, the necessity for high-quality original data, and the risks of overfitting due to aggressive augmentation techniques.

Label Preserving and Data Integrity

Ensuring that the augmented images preserve the original labels and maintain the anatomical specificity is crucial. Any changes introduced during augmentation must not alter the inherent characteristics of the anatomical structures to the extent that the labels become inaccurate. Additionally, maintaining the integrity of the data during augmentation is essential to prevent the introduction of artifacts or distortions that could mislead the training process. Augmentation techniques must be carefully designed to avoid compromising the original quality and relevance of the medical images.

Avoiding Artifacts That Compromise Diagnosis

Artifacts introduced during augmentation can compromise the diagnostic quality of the images. It is important to ensure that augmentation methods do not introduce anomalies that could be mistaken for pathological features, potentially leading to incorrect diagnoses.

Necessity for High-Quality Original Data

The effectiveness of data augmentation relies heavily on the quality of the original dataset. High-quality, accurately labeled images are essential to produce meaningful and reliable augmented data. Poor-quality original data can lead to equally poor augmented data, undermining the training process.

Overfitting Due to Aggressive Augmentation

While augmentation aims to enhance the training dataset, overly aggressive augmentation can lead to overfitting. This occurs when the model learns to recognize augmented features that do not generalize well to real-world data, thereby reducing its performance on unseen images. It is important to strike a balance in augmentation techniques to avoid this pitfall.

2.1.2 Methods for Augmenting Medical Image Data

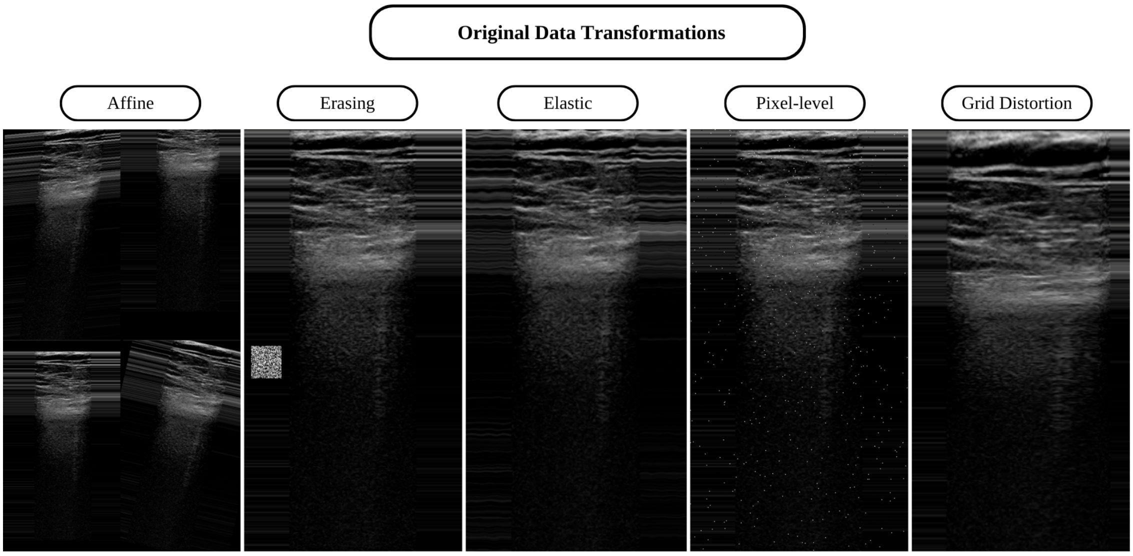


Figure 2.1: Taxonomy of Data Augmentation in medical image data

Considering the taxonomy introduced in [3], data augmentation methods in the medical domain are divided into two types: *Transformation of Original Data* and *Generation of Artificial Data*. In this study, however, we focus solely on the first type, *Transformation of Original Data*, while also incorporating *Grid Distortion* methods, thereby creating a taxonomy diagram visualized in Fig. 2.1. This diagram categorizes the different strategies that can be applied to original data, under categories such as Affine, Erasing, Elastic, Pixel-level, and Grid Distortion. Notably, random erasing is referred to as a pixel-level transform, while Elastic and Grid Distortion are considered as Spatial transforms.

Transformation of existing data involves direct modifications to the input data, enabling the simulation of a broader array of scenarios that the model might face in real-world applications.

Affine Transformations

This geometric transformation preserves lines and parallelism, but not necessarily distances and angles. It includes translation, reflection, scale, rotation, shear, and cropping transformations. All of them can be represented as matrices and hence can be combined to get composed transformations. This type is very common and is practically always a member of the data augmentation pipeline, due to ease of implementation and significant impact on model robustness [4, 5, 6, 7, 8].

Erasing Transformations

In this kind of transform, a random section of the image is replaced with random noise. This approach forces neural networks to avoid relying on simplified detection patterns and was originally designed to address issues with occlusions in RGB images – problems that are typically absent in medical images [9]. However, erasing transforms can still be beneficial for medical images by enhancing the neural network's robustness and generalization capabilities. This method is similar to Cutout [10], which uses a constant value, usually zero, instead of random noise. Additionally, this is a parameter-free technique, so there is no need for tuning [11].

Elastic Transformations

This type of augmentation involves modifying the original image through random displacement of pixels along a grid defined on the image, maintaining a smooth transition by chosen interpolation. Since they do not preserve collinearity and aspect ratio, like do-it affine transforms, they are able to create local shape variations. Based on it, it can simulate the breathing or equivalent processes inherent in medical images [12, 13, 14, 15, 16].

Grid Distortion Transformations

Grid distortion transforms involve dividing the image into a grid and then randomly displacing the points within this grid. This method creates distortions that can simulate various types of deformations that might occur during image acquisition, such as scanner or patient movement. Compared to elastic transforms, which use a smooth, continuous displacement field, grid distortion applies more abrupt, localized distortions. This technique can enhance the model's robustness to irregular deformations and improve its performance in real-world scenarios where such distortions are common [17, 18, 19].

Pixel-Level Transformations

Pixel-level transformations involve modifying the pixel values of an image to introduce invariance at the pixel level. These transformations can help neural networks become more robust to variations in the data. In the medical domain, images are often grayscale, so transformations that alter color space, like those used for RGB images, are less applicable. However, pixel-level transformations can still be valuable in medical imaging for addressing distortions that arise from different imaging protocols, scanners, and acquisition conditions [20, 21, 14, 8, 22, 23, 24].

2.2 Parallel Hyperparameter Tuning

Hyperparameter tuning is a critical component in machine learning, necessitating the selection of optimal hyperparameters for a learning algorithm [25].

The choice of hyperparameters significantly affects the behavior and efficacy of models, particularly in neural networks where settings such as the learning rate, the number of hidden layers, and the number of neurons per layer can profoundly influence the outcomes.

Manual optimization of these parameters is both time-consuming and often ineffective due to the vastness of the hyperparameter space. In response, the technique of parallel hyperparameter tuning has been developed.

The term parallel refers to the simultaneous execution of multiple tuning processes across various computational resources, such as GPUs, TPUs, and NPUs. This approach allows for the concurrent evaluation of different hyperparameter sets, significantly increasing the efficiency and scalability of the tuning process.

2.2.1 Main Approaches to Hyperparameter Tuning

A variety of approaches were designed to investigate the expansive hyperparameter space. This section provides a brief overview of the most commonly utilized methods, their advantages, limitations, and working principles.

Grid Search

This is the simplest approach, which systematically examines the defined range of hypermeters. Its main advantages are simplicity and comprehensive exploration, which ensures that every combination listed will be evaluated. However, due to this exhaustive coverage, the number of combinations will grow exponentially with adding each new hyperparameter, making it very efficient in large spaces.

Random Search

This method selects a set of hyperparameters independently for each value based on a pre-defined distribution, usually uniform. It can sometimes find near-optimal configurations faster than Grid Search in higher-dimensional hyperparameter optimization settings [26]. However, its unpredictability is also a disadvantage, as it might overlook significant combinations.

Bayesian Optimization

This technique is based on modelling the performance of configuration by constructing a probability model of the objective function and using it to select the most promising sets of hyperparameters to evaluate the true objective function. Gaussian process is commonly employed to model this objective function; it provides an estimate of the performance of hyperparameters and confidence in that estimate, which is essential for effective search space management. The evaluation is based on model outputs so poor assumptions may lead to suboptimal solutions [27].

Genetic Algorithms

Inspired by natural selection, genetic algorithms treat each hyperparameter configuration as an individual in the population. They apply operations such as crossover and mutation to evolve the population to better performance. This approach excels in navigating complex hyperparameter spaces despite being highly dependent on the calibration of operational parameters.

Gradient-Based

These methods are extremely often and successfully used to optimize neural network parameters with millions of dimensions [28]. However, in general, no gradient information is available to respect hyperparameters, so this approach is not commonly used for hyperparameter optimizations. Despite this fact, [29] have shown the possibility of training neural networks based on differentiation along the path of the best response model parameters concerning the hyperparameters.

2.2.2 Tools for Parallel Hyperparameter tuning

In modern times, various tools and platforms are available for detailed and interactive analysis of hyperparameter tuning. They offer a complex infrastructure that allows for the easy identification of optimal hyperparameters through a simple interface, while also implementing optimization algorithms to enhance the efficiency and effectiveness of the tuning process. In this section, we compare some prominent tools.

Weights & Biases

It is a robust tool extensively utilized by a broad community of researchers and engineers for the automation of machine learning experiments, as well as for tracking and visualizing results [30]. It supports parallel hyperparameter tuning through a feature known as Sweep [31]. This feature streamlines the process of evaluating and comparing the performance of various hyperparameter configurations, making it highly intuitive and user-friendly for practitioners in the field. This functionality is particularly beneficial for optimizing machine learning models efficiently, allowing users to quickly identify the most effective settings.

Ray Tune

Ray Tune is a component of the Ray framework, which is specifically designed for distributed computing [32]. This tool offers support for a diverse array of optimization algorithms, including Bayesian optimization and population-based methods. A significant advantage of Ray Tune is its scalability; it can seamlessly transition from a single machine to large clusters, accommodating heavy computational tasks effectively. However, it is worth noting that Ray Tune's complexity can present a steeper learning curve for beginner researchers, especially when compared to more user-friendly platforms like Weights & Biases.

Optuna

Optuna is an open-source hyperparameter optimization framework that stands out from other approaches due to its ability to accommodate more complex optimization logic [33]. It supports a broad range of trial schedules and pruning strategies, making it highly versatile for various types of machine learning experiments. A notable advantage of Optuna is its integration with development environments like Visual Studio Code (VSCode) [34]. This integration allows developers to seamlessly combine the coding and refinement of machine learning solutions with the visualization of hyperparameter tuning processes, all within a single application. This streamlined approach enhances productivity and simplifies the workflow, particularly beneficial for machine learning practitioners who value an efficient and integrated development environment.

2.2.3 Experiment Scheduling

For the purposes of this thesis, Weights & Biases (WandB) has been selected as the platform of choice for tracking and implementing parallel hyperparameter tuning. WandB is renowned for its user-friendly approach to automating hyperparameter searches and its provision of comprehensive, interactive data visualization capabilities. It supports three prevalent search methods: Bayesian optimization, grid search, and random search, which are integral to conducting thorough and efficient hyperparameter optimization.

Within Weights & Biases, a sweep is defined as a series of experiments orchestrated to systematically explore various hyperparameter configurations. These sweeps are coordinated by a Sweep Controller, which operates from the platform's cloud infrastructure. This controller is responsible for generating a set of directives based on the user-provided configuration, which includes the chosen method for navigating the hyperparameter search space, the metric to be optimized, and the specific ranges for each hyperparameter.

These directives are subsequently executed by agents, which are components designated to carry out the experimental runs. These agents are equipped with functionalities to pause, resume, stop, and cancel runs as required, providing flexibility and control over the tuning process. This structured approach ensures a systematic and efficient exploration of the hyperparameter space.

2.3 Neural Network Architecture Selection

This thesis explores the impact of data augmentation methods on ultrasound image classification, necessitating the careful selection of a suitable neural network architecture. Among the multiple award-winning architectures evaluated, including VGGNet, GoogLeNet, LeNet-5, and ResNet, the choice was narrowed to ResNet, or Residual Neural Network. Developed by researchers at Microsoft, ResNet distinguished itself significantly by securing first place at the ILSVRC 2015 classification task [35].

ResNet is particularly notable for its innovative use of residual connections. These connections are a strategic response to the degradation problem often observed as the neural network depth increases [35].

The fundamental component of ResNet is the residual block, which is designed to learn specific transformations on the input data. To delve into the mechanics, envision a typical residual block comprising two sequential layers Fig. 2.2. Here's a breakdown of the operations within the block:

1. **First Layer**: The initial layer receives an input x and applies a transformation F_1. This transformation typically encompasses several operations including convolution, batch normalization, and activation via the ReLU function. Consequently, the output from this layer is an intermediate value given by:

$$u = \mathrm{ReLU}(F_1(x)) \tag{2.1}$$

2. **Second Layer**: Subsequent to the first layer, the second layer acts upon the intermediate output u to perform another transformation, F_2, resulting in:

$$v = F_2(u) \tag{2.2}$$

However, the essence of a residual block lies in its approach to handle this output. Rather than accepting v as the final output, the block calculates a residual. The residual output $F(x)$ is computed by adding v to the original input x, facilitated by a shortcut connection that bypasses the two transformational layers. The equation is as follows:

$$F(x) = v + x \tag{2.3}$$

3. **Activation**: The final operation in the residual block involves another ReLU function applied to $F(x)$, culminating in the block's output:

$$y = \text{ReLU}(F(x)) \tag{2.4}$$

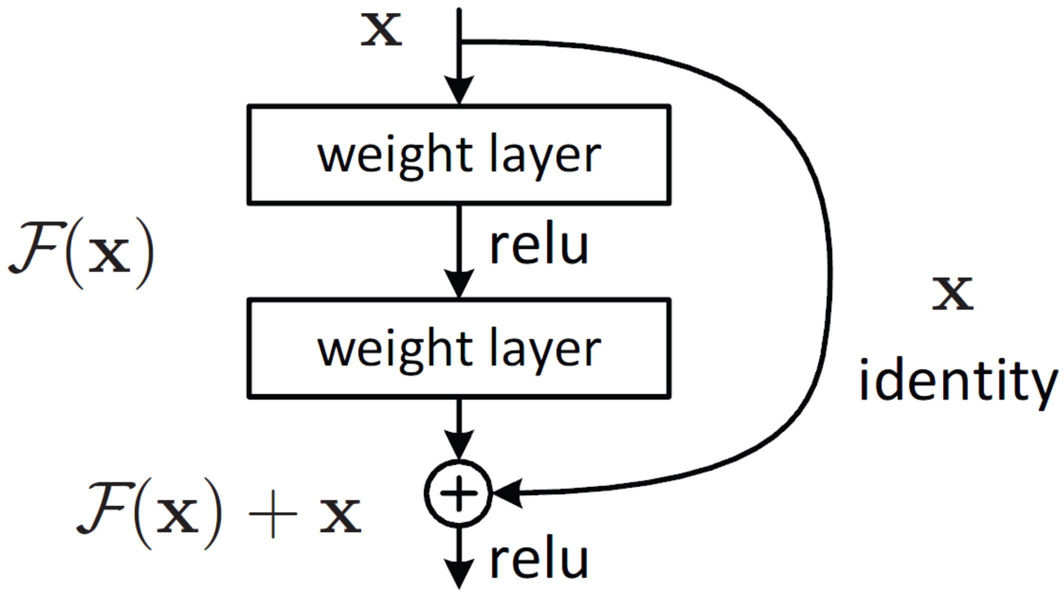


Figure 2.2: Illustration of a Residual Block in ResNet Architecture [35]

The fundamental concept underlying the ResNet architecture is residual learning, which is pivotal in its ability to train very deep neural networks effectively. In traditional neural network architectures, each layer is tasked with learning the direct transformation from input x to output y. However, ResNet modifies this approach by enabling each layer within a residual block to focus on learning the residual [35]:

$$R(x) = F(x) - x \tag{2.5}$$

The residual learning framework addresses the challenge of vanishing gradients, which is a prevalent issue in training deep neural networks. As the neural network depth increases, gradients passed back during training can become exceedingly small, effectively halting the neural network from learning further—this hampers the training process and degrades the neural network performance. Residual blocks in ResNet allow for a direct flow of gradients through shortcut connections that skip one or more layers, thereby maintaining the gradient's strength across the neural network's depth [35].

3 Synthetic chapter

In this chapter, we describe the dataset used in our experiments, detailing their characteristics and the preprocessing steps undertaken.

We will outline the Bedside Lung Ultrasound in Emergency (BLUE) Protocol and its relevance to our study, discuss different types of ultrasound probes, and elaborate on the dataset's preprocessing and cross-validation techniques.

This chapter sets the stage for the practical implementation and evaluation of data augmentation methods.

3.1 Dataset

In the realm of healthcare, particularly in diagnosing pulmonary diseases, various imaging modalities are employed to detect and analyze lung conditions. These include chest X-rays, computed tomography scans, magnetic resonance imaging, positron emission tomography, sputum smear microscopy images, and molecular imaging. Among these, X-rays and computed tomography scans are the most prevalently used anatomic imaging techniques for identifying and diagnosing a range of lung diseases.

However, a significant development occurred in 2008 [36] with the introduction of the Bedside Lung Ultrasound in Emergency (BLUE) Protocol, which aimed to facilitate rapid diagnosis in patients experiencing acute respiratory failure. This protocol leverages lung ultrasonography and categorizes findings into specific protocol profiles, which, according to the study, would have yielded correct diagnoses in 90.5% of cases. The BLUE Protocol is shown in Figure 3.1.

The BLUE Protocol underscores the growing relevance of lung ultrasound as a critical diagnostic tool in emergency and respiratory medicine, providing a quick, non-invasive, and effective method for assessing pulmonary conditions directly at the bedside.

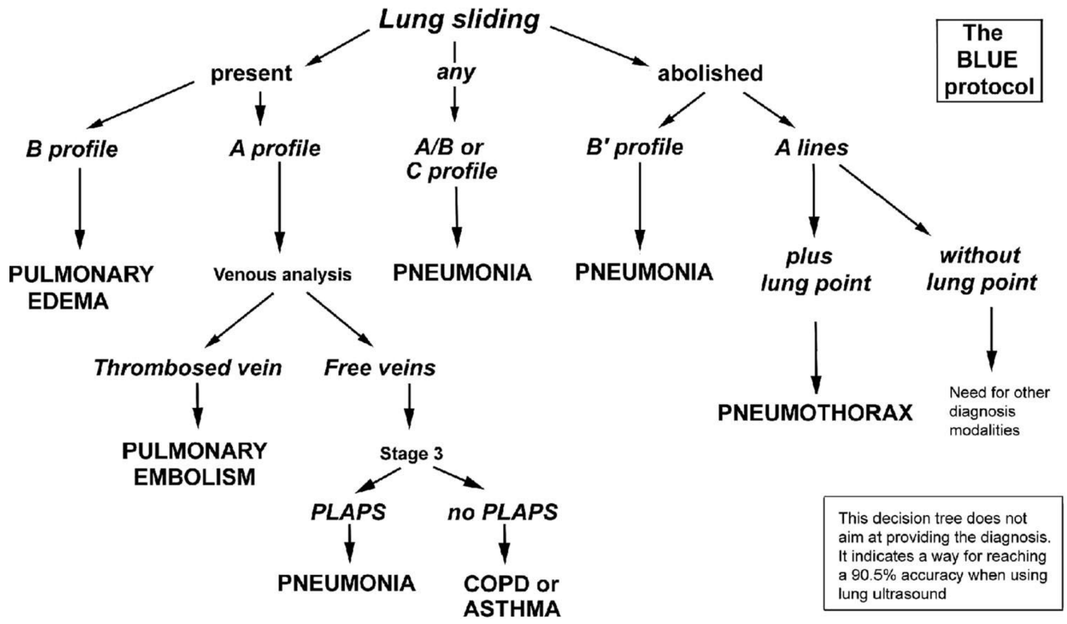


Figure 3.1: The Bedside Lung Ultrasound in Emergency (BLUE) Protocol [36]

3.1.1 BLUE Protocol and Its Decision-Making Profiles

The BLUE (Bedside Lung Ultrasound in Emergency) Protocol is structured as a decision tree, where the leaf elements represent specific diseases, and each decision node corresponds to the presence or absence of particular visual artifacts observed on lung ultrasonography images. By following the paths in the decision tree, healthcare providers can efficiently identify lung conditions with a high degree of accuracy, making this protocol an invaluable tool in the rapid assessment of patients with acute respiratory failure

Lung sliding

Lung sliding is a sonographic artifact that signifies normal respiratory motion in lung ultrasonography. It is visualized as a shimmering or sliding movement of the pleural line, which is the bright, echogenic line visible at the interface between the lung and the chest wall, occurring as the patient breathes. The presence of lung sliding is generally indicative of the absence of pneumothorax at the examined site of the thorax. This is because a pneumothorax would interrupt the normal sliding motion of the pleural layers against each other, due to the presence of air in the pleural space.

A Lines

A lines are horizontal, repetitive echoes that appear beneath the pleural line. They are reverberation artifacts, meaning they are created by the ultrasound waves bouncing between the probe and a highly reflective surface, such as the pleura. A lines are equidistant from each other and the pleural line. The presence of A lines typically indicates that there is normal air content in the lungs or possibly an over-inflated lung, but no fluid or consolidation. An example of A lines is shown in Fig. 3.2(a).

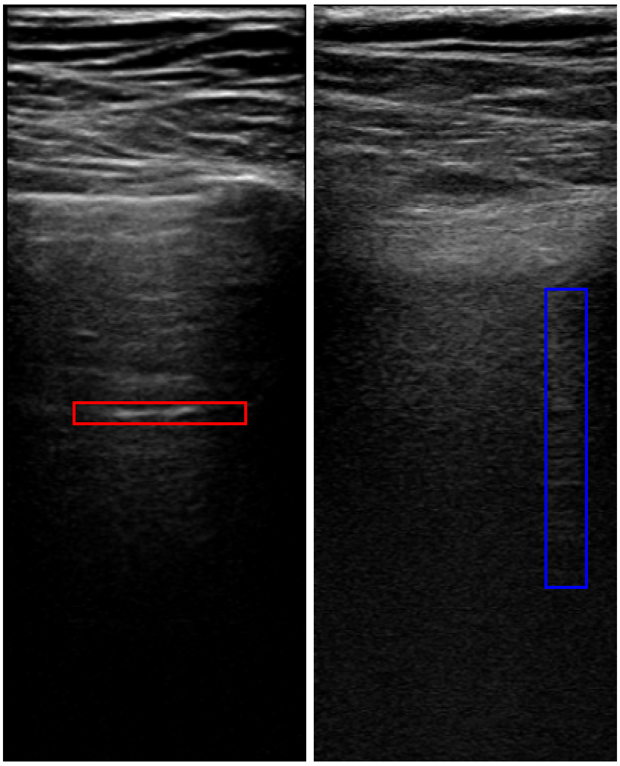
B Lines

B lines, also referred to as lung comets, are vertical, hyperechoic artifacts that project from the pleural line to the edge of the ultrasound screen, and they oscillate in sync with lung sliding. These artifacts signify an increased lung density, typically resulting from conditions such as edema, inflammation, or fibrosis. The detection and quantity of B lines on an ultrasound can aid in diagnosing various pulmonary issues, including pulmonary edema, interstitial lung disease, and pneumonia. An example of B lines is shown in Fig. 3.2(b).

3.1.2 Types of Ultrasound Probes

The final lung ultrasonography image is produced by emitting high-frequency sound waves, which range from 2 to 15 MHz, from a probe directed at the lung area. These sound waves are reflected back to the probe and then converted by a computer into a two-dimensional ultrasonography image

To cater to different diagnostic needs, various types of ultrasound probes are employed, each designed for specific applications and anatomical considerations. These probes vary in their frequency, size, and the depth of penetration they offer, allowing clinicians to choose the most suitable probe based on the specific requirements of the examination (see Fig. 3.3).



(a) A lines (red) box (b) B lines (blue box)

Figure 3.2: Examples of lung ultrasonography artifacts

Linear probe

The linear ultrasound probe is a high-frequency transducer, typically ranging from 5 to 15 MHz, renowned for its ability to provide the highest resolution among ultrasound probes. However, its use is limited to visualizing superficial structures due to its shallow penetration depth. A practical guideline is that the linear probe should be selected for imaging targets that are less than about 8 cm deep; beyond this depth, its effectiveness diminishes significantly. This probe offers a rectangular field of view, which directly corresponds to its linear footprint.

Curvilinear probe

The curvilinear ultrasound probe operates within a frequency range of 2 to 5 MHz, categorizing it as a low-frequency probe. It features a large or wide footprint, which enhances its lateral resolution relative to that of the phased array probe. This type of probe is predominantly utilized for abdominal and pelvic ultrasound examinations due to its ability to penetrate deeper into the body's tissues, capturing broader images necessary for these applications. It is limited by the large footprint and difficulty with scanning between rib spaces.

Phased array probe

The phased array (or sector array) transducer is often referred to as the "cardiac probe" due to its primary application in cardiac imaging. It operates within a frequency range of 1 to 5 MHz, similar to the curvilinear probe, but distinguishes itself with a smaller and flat footprint. This compact design is particularly advantageous for cardiac examinations because the piezoelectric crystals, which are the active elements within the probe, are densely layered and centrally packed. This configuration facilitates easier access to confined spaces, such as between the ribs, where larger probes might struggle to fit.

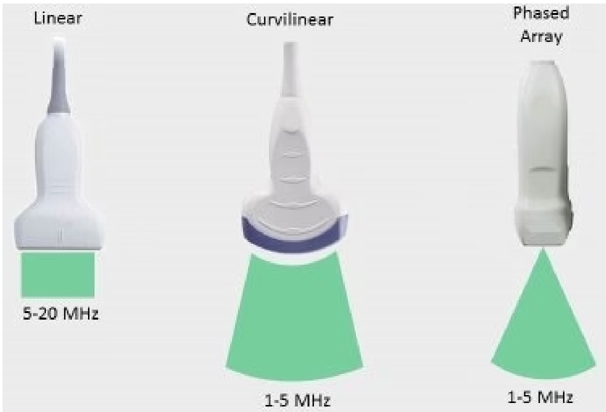


Figure 3.3: Types of ultrasound probes [37]

3.1.3 Data acquisition and character of dataset

The dataset used for training the neural network was obtained directly from the Department of Thoracic Surgery and the Clinic of Radiology at Jessenius Faculty of Medicine in Martin. The dataset was compiled through the collaboration of medical professionals from this institution, ensuring that the images were captured by experts in the field. Ethical considerations were meticulously adhered to, recognizing the sensitive nature of medical data. All images were captured using the Philips Lumify linear ultrasound probe, renowned for its high-resolution imaging capabilities.

In the given dataset, there were instances with different variants of artifact presence. For example, a lung could exhibit B lines with or without lung sliding. Similarly, A lines could also be present with or without lung sliding. However, in our experiments, we focused on improving the accuracy of classifying the presence or absence of B lines. Thus, for the purpose of training, the final labels were defined as follows: 1 for the presence of B lines and 0 for the absence of B lines.

Preprocessing

In our image preprocessing pipeline, due to the variability in the sizes of the images within the dataset, we first resize each image to a specific target size. We opted to resize the images to 600x400 pixels (Height x Width), intentionally choosing a non-square dimension to closely align with the approximate proportions of the raw images. This resizing process maintains the original aspect ratio and employs antialiasing techniques to enhance the image quality.

Following the resizing, the images are padded to match a predefined rectangular size, ensuring they are centrally aligned. The padding is applied symmetrically using an 'edge' mode that extends the borders of the images seamlessly. This approach standardizes the dimensions across the dataset, facilitating consistent handling during the training process.

Cross-Validation

To enhance the robustness of our evaluation further, we implement a 5-fold cross-validation technique. In this approach, the dataset is divided into five subsets. During each fold of the validation process, a different subset is used as the test set while the remaining four subsets are combined to form the training set. This method is particularly advantageous in scenarios with limited data, such as ours. It ensures that every data point is used for both training and validation, allowing for a comprehensive assessment of the model's performance across various subsets of the data.

Dataset Distribution for Each Fold

The following table (3.1) demonstrates how the dataset is distributed across each fold, including the divisions into training, validation, and test sets. The table provides a detailed view of the number of positive and negative instances in each subset. Here, positive instances refer to samples with B lines present, while negative instances refer to samples without B lines.

Fold	Subset	Total Positives (B lines present)	Total Negatives (B lines absent)
1	train	111	195
	test	27	203
	val	27	50
2	train	112	196
	test	29	301
	val	24	53
3	train	119	190
	test	27	144
	val	19	59
4	train	96	165
	test	47	166
	val	22	44
5	train	107	181
	test	35	215
	val	23	49

Table 3.1: Distribution of positive (B lines present) and negative (B lines absent) samples for each fold

3.2 Single methods to be evaluated

As in theoretical side we defined a general description of commonly used data augmentation strategies for medical domain, in this section we wil define concrete transformation methos along with their implementing specifity and parameter limits. Each single method will be evaluated as the first part of the experiments.

3.2.1 Affine and Spatial Transformations

Firstly, we start with designing methods from the simplest type and most commonly used in the medical domain as data augmentation. In this section, we have also included Grid Distortion and Elastic Transformation, which were previously discussed in the theoretical part. Here are the methods:

Translation

In this process, we apply random horizontal and vertical translations to an image, controlled by two parameters: a and b. The parameter a governs the horizontal shift and is randomly selected from a range defined as $[-\text{image width} \times a, \text{image width} \times a]$, where image width is the width of the image. Similarly, b manages the vertical shift, with values randomly chosen from the interval $[-\text{image height} \times b, \text{image height} \times b]$, where image height is the height of the image. We explore four distinct parameter configurations for these translations: [(0.15, 0.0), (0.0, 0.15), (0.15, 0.15), (0.3, 0.3)].

Rotation

Rotation of an image is achieved by defining a range for both clockwise and counter-clockwise rotations. Positive values represent clockwise rotations, and negative values represent counter-clockwise rotations. The specific rotation angle is chosen randomly within the specified range. The degrees of rotation tested include ranges of [(-5, 5), (-15, 15), (-30, 30), (-45, 45)].

Shear

The shear transformation slants an image along the x-axis and y-axis, controlled by four parameters that determine the shear range. These parameters are randomly selected from sets including [(-15, 15, 0, 0), (0, 0, -15, 15), (-15, 15, -15, 15), (-30, 30, -30, 30)], allowing for varied horizontal and vertical adjustments.

Reflection

In the context of medical imaging, vertical flipping is generally not applicable due to the nature of the anatomical structures. Therefore, our experiments are limited to horizontal flipping, where the image is mirrored along the vertical axis. This transformation helps simulate variations that may be encountered in real-world medical data.

Scale and Crop

Scaling involves resizing the image by a factor that is randomly chosen from a specified interval while preserving the original dimensions. Cropping adjusts the image by focusing on a portion of the original frame and then resizing the cropped area back to the original dimensions. Parameters for these methods include sets such as [(1.0, 1.15), (0.85, 1.0), (0.85, 1.15), (0.7, 1.3)] for scaling and [(0.7, 1.3), (0.85, 1.15), (0.85, 1.0), (1.0, 1.15)] for cropping.

Grid Distortion

This method employs two parameters: the number of cells into which the image is divided—both horizontally and vertically—and the distortion magnitude for each cell segment. We conducted tests with various grid setups, including [9, 7, 5, 3] cells, and distortion magnitudes ranging from 0.03 to 0.5, resulting in 20 distinct experimental conditions.

Elastic Transformation

This technique uses two parameters: alpha, which controls the strength of the displacement, and sigma, which affects the smoothness of the displacement field. We evaluated this method using alpha settings [(5, 50), (5, 40)] and sigma in the interval (4.0, 5.0), as well as [(5, 10), (5, 20)] with sigma (3.0, 5.0), to simulate a range of elastic effects. It is important to note that both alpha and sigma are chosen randomly within their respective intervals for each transformation.

3.2.2 Pixel-level Transformations

Pixel-level transformations focus on manipulating individual pixels to simulate various real-world imaging conditions, particularly useful in medical imaging where noise can significantly affect diagnosis. This section also includes Random Erasing as a pixel-level transformation method.

Salt and Pepper Noise

Salt and Pepper noise randomly alters pixels to their maximum (salt) or minimum (pepper) values, mimicking the appearance of random black and white specks on the image. This noise type is parameterized by its density, which specifies the proportion of the image's pixels that will be affected. For each application of this noise, the density is randomly selected from the provided intervals or set to a fixed value. We evaluated this noise with densities in intervals [(0.005, 0.01), (0.01, 0.03)] and fixed values [0.001, 0.01], resulting in four distinct transformations

Gaussian Noise

Gaussian noise, or normal noise, is implemented using parameters that define the mean and standard deviation of the Gaussian noise distribution. This type of noise adds a Gaussian-distributed pixel intensity variation across the image. The noise parameters tested included a mean of 0 and standard deviations [0.01, 0.05, 0.1, 0.5], encompassing four variations.

Noise with Fixed Value

This method, designed to test a specific noise condition, uses a fixed noise value that varies across different tests. The ranges for this noise parameter were [(0.1, 0.5), (0.1, 1.0), (0.5, 1.0), (0.5, 1.5), (1.0, 1.5)] and fixed values [0.1, 0.5], adding up to seven possible settings. This method designed at [38].

Speckle Noise

Unlike Gaussian noise that adds values, Speckle noise multiplies the image by a Gaussian noise matrix, affecting the image in a way that simulates speckles commonly seen in ultrasound images. It was tested with mean values set to 0 and standard deviations [0.05, 0.5, 0.1, 0.8], creating four different conditions.

Brightness Adjustment by addition

This method involves adding a randomly selected value from a specified interval to the pixel intensities of the image, effectively altering its overall brightness. The parameter—delta interval—defines the range of values that can be added to or subtracted from the image's brightness. We have tested this method with the following intervals: [(0.005, 0.015), (0.15, 0.25), (0.25, 0.35), (0.05, 0.15), (-0.15, 0.15), (-0.25, 0.25), (-0.35, 0.35)], resulting in seven distinct settings. This method designed at [38].

Brightness Jitter

This method adjusts the brightness of an image by blending the original image with a completely black image. The extent of this blending is controlled by a randomly chosen ratio, which ranges from -0.3 to 0.3. This ratio indicates how much of the black image is mixed with the original. A negative ratio slightly darkens the image, while a positive ratio lightens it. Essentially, the closer the ratio is to zero, the less change to the brightness; moving away from zero increases the effect, either brightening or darkening the image depending on the direction of the ratio.

Contrast Jitter

This method modifies the contrast of an image by blending the original image with a version of itself where all pixels have been set to the average pixel intensity of the original image. This blending is governed by a randomly selected ratio within ranges like [(-0.1, 0.1), (-0.3, 0.3), (-0.5, 0.5)].

Combined Brightness and Contrast Jitter

This technique simultaneously adjusts both brightness and contrast in a single operation. It uses a similar approach to the individual jitter methods by blending the original image with modified versions—either darker, lighter, or averaged based on the mean pixel intensity. The extent of the adjustment is controlled by selected ratios from ranges such as [(-0.1, 0.1), (-0.15, 0.15), (-0.3, 0.3), (-0.4, 0.4)].

Contrast by Multiplication

This function directly alters the contrast of an image by multiplying each pixel value by a predetermined factor. The multiplier is randomly chosen from a set range such as [(0.5, 1.5), (0.1, 0.5), (0.01, 0.1), (0.5, 1.0), (1.5, 2.0), (0.9, 1.1)]. This method designed at [38].

Adjust Contrast by Smooth Field

This advanced method adjusts the contrast by calculating a smooth field over the image. First, the image is divided into a grid specified by parameters, such as a 2x2, 3x3, or 5x5 grid. Each cell in this grid has its contrast adjusted by a factor derived from a 'gamma' range, such as [(0.5, 1.5), (0.5, 4.5)]. The gamma parameter controls the exponential factor applied to each cell, thereby adjusting the luminosity of the pixels within each segment of the grid.

Gaussian Blur

Gaussian Blur is applied to an image to reduce detail and noise. It works by convolving the image with a Gaussian function; essentially, each pixel is transformed to become the weighted average of itself and its neighbors. The amount of blur is controlled by two parameters: kernel size and sigma. The kernel size specifies the size of the neighborhood over which the average is taken, and sigma, which influences the shape of the Gaussian function, determines how much weight is given to pixels further away from the central pixel. The ranges tested include sigma values like [(0.1, 0.5), (0.5, 1.0), (1.5, 2.0)]. For the kernel size, we tested values [3, 5], resulting in a total of 6 configurations.

Gaussian Sharpening

Contrary to Gaussian Blur, Gaussian Sharpening enhances the edges and details within an image. This process involves first applying a Gaussian blur to the image using a broader sigma, blurring the image significantly. A second, narrower Gaussian blur is then applied to the first blurred image. The sharpened image is produced by adding the weighted difference between these two blurred images back to the original blurred image, scaled by a factor called 'alpha'. The parameters, including sigma and alpha, have been experimentally determined with limits such as sigma (0.5, 1.0) and alpha [(1.0, 5.0), (0.0, 2.0)]. Here, the chosen parameters are randomly selected from the specified ranges.

Adjust Sharpness

This transformation adjusts the sharpness of an image by blending it with a slightly blurred version of itself. The factor determining the extent of this blending ranges from 0.0, yielding a completely blurred image, to 1.0, preserving the original sharpness, and values greater than 1.0, which enhance sharpness. For instance, if the factor is set at 2.0, the image sharpness is increased by a factor of 2, making it crisper and more defined. For experimental purposes, we tested this method with four distinct sharpness factors: [1.5, 2.0, 2.5, 3.0].

Shift Intensity

This transformation method involves shifting the intensity of an image by applying a randomly selected offset to each pixel's value. The offset is randomly chosen from specified intervals or a fixed value, altering the overall brightness or darkness of the image. The intervals we experimented with for this method are [(-0.1, 0.1), (-0.1, 0.4), (0.1, 0.4)], along with fixed values [0.05, 0.1].

Random Erasing

Random Erasing is a transformation technique that selectively obscures a region of an image by erasing its content. This method involves two parameters: the scale of the erased region relative to the image size, and the replacement value used to fill the erased area. The scale parameter, which defines the size of the erased region, was tested within a range from 0.001 to 0.01 of the image size, ensuring that only a small portion of the image is affected.

For the replacement value, we employed two approaches: one where the erased pixels are set to zero, rendering them completely black, and another where the pixels are filled with random values, introducing noise.

3.3 Model Training and Evaluation Techniques

Given the limited size of our dataset—a common scenario in the medical domain due to factors such as the scarcity of labeled medical images, high costs associated with data annotation, and privacy concerns—training a neural network from scratch using this dataset alone is suboptimal. Training from scratch in such circumstances would likely lead the model to focus on learning pixel-specific details of the images rather than extracting generalizable patterns across classes.

This situation warrants the use of transfer learning [39], a technique where a model is initially trained on a comprehensive image dataset and subsequently fine-tuned on a smaller, domain-specific dataset. This approach leverages pretrained models to learn general characteristics and critically important structural features of images, including low and mid-level patterns. For our purposes, we utilized the standard pretrained ResNet-18 model, originally trained on the extensive ImageNet dataset. ResNet models are available in various configurations (e.g., 18, 36 layers), but ResNet-18 was selected due to its fewer parameters, which mitigates the risk of overfitting given the small size of our dataset and maintains a balance between model complexity and accuracy.

3.3.1 Training Process and Hyperparameters

For training, we selected cross-entropy loss as the criterion and Adam as the optimizer [40]. While different choices of models, criteria, or optimizers might slightly alter the classification accuracy, these components can be easily adjusted in the configuration file. The primary focus of this work is on exploring the impact of various augmentation methods rather than optimizing these other model components. This approach allows for a more concentrated examination of how augmentations can enhance model training and performance under constrained data conditions.

The hyperparameter determining the number of epochs—how many times the entire training set is passed through the network—was set to 40. This number was deemed sufficient for the model to familiarize itself with the new dataset and learn its unique characteristics thoroughly.

Furthermore, we selected a batch size of 32 for training. This size was optimized so that each batch, along with the model, can fit on a single GPU, enhancing our computational efficiency. The learning rate was set to 0.0001 to facilitate a steady reduction in the training loss, promoting effective learning without overshooting the minimal loss values.

Early stopping

Additionally, to ensure the model does not overfit and to monitor the generalization error effectively, we implemented early stopping as a regularization method. This technique involves setting a patience parameter, which in our case was chosen as 5. The early stopping mechanism monitors the validation error: if there is no decrease in this error for five consecutive epochs, the training process is halted. This approach helps in preventing the model from over-training on the dataset, thus optimizing computational resources.

Data Augmentation Strategy

Data augmentation plays a pivotal role in our training approach, significantly enhancing the robustness and generalization capability of the neural network. In our methodology, data augmentation is applied in an online (on-the-fly) manner. This technique involves dynamically transforming images during the training process, ensuring that each batch is augmented uniquely and introducing a diverse range of variations to the training data.

3.3.2 Evaluation Metrics

To rigorously assess the performance of the model in the context of binary classification, we rely on a set of standard evaluation metrics derived from the confusion matrix and further calculations of precision, recall, and the F1-score.

The confusion matrix is a fundamental tool in classification tasks, providing a visual representation of the performance of a classification algorithm. For binary classification, the matrix includes four key elements:

- True Positives (TP): The number of positive instances correctly classified as positive.
- True Negatives (TN): The number of negative instances correctly classified as negative.
- False Positives (FP): The number of negative instances incorrectly classified as positive.
- False Negatives (FN): The number of positive instances incorrectly classified as negative.

Derived from these four foundational elements, several key metrics provide insight into the accuracy and efficacy of the model:

Precision

Precision is a measure of the accuracy of the positive predictions made by the model. It is calculated using the formula:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.1}$$

This metric indicates the proportion of positive identifications that were actually correct and is particularly useful when the costs of False Positives are high.

Recall

Recall, also known as sensitivity, measures the ability of the model to identify all relevant instances. For binary classification, it is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.2}$$

This metric is crucial in scenarios where missing a positive instance (such as failing to identify a disease) has serious ramifications.

F1-Score

The F1-score is the harmonic mean of precision and recall, providing a single score that balances both the precision and the recall. It is particularly useful when you need to balance the importance of precision and recall. The F1-score is calculated as follows:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.3}$$

This metric is crucial when there is a need to balance precision and recall, especially in the presence of imbalanced datasets where one class may predominate. This dominance can render a high accuracy score misleading, highlighting the importance of a balanced metric like the F1-score.

Collectively, these metrics furnish a comprehensive evaluation of the model's performance, ensuring not just high accuracy, but also an equilibrium in detecting positives and correctly identifying negatives within binary classification tasks.

K-Fold Metrics

Given that our experiments utilize k-fold cross-validation, we aim to objectively assess the performance of the model with each selected augmentation configuration across all k distributions of the dataset. To achieve this, we calculate the mean, maximum, minimum, and standard deviation (std) values for each metric across all folds. These statistics are computed for each set of runs grouped by the applied augmentation pipeline. The inclusion of the standard deviation helps in understanding the variability and consistency of the model's performance across different folds.

4 Evaluation

In this chapter, we thoroughly assess the effectiveness of various data augmentation techniques on the performance of a neural network trained for classifying lung ultrasound images with the presence or absence of B-lines. The evaluation is divided into two main sections: the first focuses on the impact of single data augmentation methods, while the second explores the effectiveness of combined augmentation strategies.

The primary objective is to determine how different augmentation methods and their combinations influence the model's ability to generalize from the training data to unseen test data. This assessment is crucial in ensuring that the model performs robustly and accurately in real-world clinical settings, where data variability is a common challenge.

4.1 Impact of Single Data Augmentation Methods

Within this section, we aim to assess the impact of various single data augmentation methods, each defined with specific parameters as outlined in section 3.2, on the performance of our model. To facilitate a clear comparison, for each type of augmentation, we also maintain one configuration devoid of any augmentation methods—relying solely on the preprocessing steps discussed in section 3.1.3.

Given that our evaluation employs a 5-fold cross-validation approach, each configuration does not undergo a single run but is instead tested across five different iterations. This setup includes the same augmentation method or the baseline with no augmentation, yet with varying distributions of the dataset. Considering the number of experiments with affine transforms totaling 21, elastic transforms 24, pixel-level manipulations 63, and random erasing 2, we have 110 individual runs involving a single data augmentation method, each applied with a

50% probability. In addition, there are 5 runs that only incorporate preprocessing transformations. As previously mentioned, each run is replicated 5 times; thus, the total number of runs for testing amounts to 575. This extensive testing framework ensures comprehensive assessment and robust validation of the augmentation methods' effectiveness.

Before analyzing the results of each augmentation type, we consider the classification accuracy for models trained without augmentations, as shown in Table 4.1. The F-1 mean varies from 0.149733 to 0.249238, indicating that performance without augmentations is somewhat varied, with significant differences between the runs. This suggests a level of inconsistency in the performance of models trained without augmentations.

Notably, runs 3, 4, and 5, which constitute the majority of the experiments, have an F-1 mean lower than that of all experiments with applied augmentations under the chosen best parameters. Additionally, only one run (run 2) managed to achieve an F1-Mean above 0.70, whereas in experiments with augmentations, 46 out of 110 attempts surpassed this threshold, with half of them exceeding the 0.80 mark.

Run	Test F1 Mean	Test F1 Std
1	0.249238	0.229130
2	0.235569	0.300148
3	0.174774	0.197802
4	0.149733	0.179493
5	0.209870	0.215907

Table 4.1: Results for runs without augmentations

4.1.1 Overall analysis

After training the models, we can broadly analyze the effect of applying individual data augmentation methods on the performance of the test set by examining Fig. 4.1. Each point on the plot represents a specific configuration, encompassing the augmentation method and its parameters, grouped based on fold distributions. Different colors represent different types of augmentation methods.

Figure 4.1: Impact of Data Augmentation Methods on Test Performance

Baseline Comparison

The cyan dots, representing runs without any augmentation, serve as a baseline for evaluating the effectiveness of augmentation methods. When discussing the advantages, it is evident that these dots often cluster in the center of the graph, indicating that some models using augmentation methods experience a negative impact during training. However, there is also a portion of these methods that enhance the classification ability of the model, thanks to more suitable techniques or parameters.

On the downside, it is clear that certain models employing Affine, Pixel-level, and Grid Distortion transforms significantly improve F-1 scores compared to models without augmentations.

Performance Spread

Each augmentation method spreads across a range of mean F1-scores, which indicates variability in how each method affects model training and generalization. Notably, some methods like Pixel-Level Transformations and Grid Distortion show data points both at lower and higher ends of performance, suggesting that the impact of these methods may be highly dependent on specific parameters or image characteristics.

4.1.2 Affine and Spatial Transformations

Referring specifically to affine transformations, these intuitive and simple methods have shown promising results, demonstrating that for all methods with the best parameters, the F-1 mean value is significantly better than the best run without any augmentation (see Table 4.2). This indicates that the use of these methods almost always contributes positively to the model's performance. The Reflection method, which does not have parameters and whose influence is not dependent on the chosen parameter, showed good results for both F-1 mean and F-1 max, with values of 0.349859 and 0.810127, respectively. This suggests that this method is well-suited for our data.

Next, the Rotate and Translate methods demonstrated strong performance. The Rotate method, regardless of the chosen parameter, showed results better than any run without augmentation. With well-chosen parameters, it showed significant improvement in the model's performance, as evidenced by the F1-mean and F1-max metrics of 0.365285 and 0.843373, respectively. Compared to the baseline model, the F-1 mean increased by more than 0.10. The Translate method also provided notable improvements, though slightly less than Rotate, but exhibited a more robust distribution relative to the chosen parameters.

The Crop and Scale methods did not have as strong an effect on performance compared to the previous two methods but still improved the F-1 mean relative to the best baseline model. With optimal parameters, the Shear method exceeded the 0.30 mark for F-1 mean, achieving 0.307217, and showed a good F-1 max of 0.833333. However, with suboptimal parameters, it can yield poorer results, though still better than most experiments without augmentation or methods like Rotate or Flip.

For the Grid Distortion method, the best-selected parameter achieved a very high F1-max of 0.946237, indicating that it introduced changes that added some invariance to the training dataset, facilitating more robust classification. Overall, among all 20 experimented configurations, methods with parameters of num cells 5, 7, and 9 and a relatively small distortion limit provided the highest stability in F1-means, ranging from 0.239087 to 0.274354, suggesting these methods positively impact our dataset. Similarly, Elastic transformation, like Grid Distortion, significantly improved our dataset, with greater improvement observed with larger parameter selections, such as at least (5.0-20.0), yielding F1-means ranging from 0.219231 to 0.268857.

Method	Best Parameter	Test F1 Mean	Test F1 Std
Rotate	(-30, 30)	0.365285	0.299338
Scale	(0.85, 1.15)	0.287642	0.291927
Shear	(-15, 15, -15, 15)	0.307217	0.309257
Translate	(0.15, 0.15)	0.350255	0.334019
Crop	(1.0, 1.15)	0.283542	0.260089
Flip	horizontal	0.349859	0.342698
Grid Distortion	(9, (-0.09, 0.09))	0.274354	0.232933
Elastic Transform	((5, 50), (4, 5))	0.268857	0.177985

Table 4.2: Results for affine and spatial transformations

4.1.3 Pixel-level Transformation

Considering the results for Noise with Fixed Value, all runs with these method had an f1-mean lower than the best baseline. For salt and pepper noise, the results were not as clear-cut. Among the three closely valued parameters, one achieved an f1-mean of 0.303765 and an f1-max of 0.303765, while the other two had f1-means of approximately 0.15 and f1-max values around 0.50. This indicates a significant variation in performance depending on the specific noise parameter applied (see Table 4.3).

Speckle noise demonstrated more stable results across different parameters, with f1-means ranging from 0.19781 to 0.243307, which is better compared to all baseline runs and similar to the best baseline run. Gaussian noise showed results lower than the best baseline model, with f1-means ranging from 0.151506 to 0.246527. It is worth noting that at least one run for each noise type experienced very low f1-mean values (Noise with FV: 0.091463, Speckle noise: 0.064286, 0.010811). This is likely due to the use of excessively large parameters, making the noise too aggressive and potentially distorting or completely removing important classification information, such as artifact structures.

Next, we evaluated brightness, contrast, and sharpness adjustments. The brightness adjustment by addition method, even with non-optimally selected parameters, was relatively stable (compared to all baseline runs), with f1-means ranging from 0.215564 to 0.333553, and one run with 0.177327. Notably, when the parameter included negative values (e.g., (-0.15, 0.15)), the f1-means were 0.215564, 0.231989, and 0.333553.

Contrast by multiplication showed that only half of the experiments achieved an f1-mean close to the best baseline f1-mean, and two even slightly exceeded the best baseline f1-mean. This suggests that this method, with the best parameters, improved the f1-mean by only 0.047 compared to the best baseline f1-mean. For ColorJitter, the results indicate that with relatively small selected parameters (up to 0.3 for contrast or the combination of contrast and brightness), the f1-means ranged from 0.223082 to 0.282096, which is quite effective in increasing the minimum f1-mean.

Adjust Contrast by Smooth Field demonstrated that only one of the first parameters produced good results, with f1-means ranging from 0.210342 to 0.315087. The other two methods yielded lower results, 0.149126 to 0.212566 respectively. This method should be explored further for the influence of the first parameter, as well-chosen values can improve stability (min f1) compared to baseline runs.

Gaussian Sharpening and Adjust Sharpness showed f1-means of 0.229106-0.254306 and 0.202591-0.269351 among all experimented parameters, respectively. These methods did not significantly increase the f1-mean compared to the best baseline model run but positively influenced the increase and stabilization of the minimum f1 value, maintaining at least the 0.20 mark.

The Erasing method was experimented with using only two parameters: random or zero replacing values. It resulted in f1-means of 0.254103 and 0.197842 and f1-max values of 0.829268 and 0.588235, indicating that this approach is beneficial with either parameter, though random replacing shows more improvement. Further improvement might be possible by erasing more than one region of the image.

The final two methods, Shift Intensity and Gaussian Blur, both showed stable improvements in f1-means to about 0.20. In the first method, the best parameter achieved an f1-mean of 0.311274 and an f1-max of 0.839506. In the second method, the f1-mean was 0.259813, and the f1-max was 0.853659, indicating these methods enhance classification accuracy but not as significantly as some other top-performing methods.

Method	Best Parameter	Test F1 Mean	Test F1 Std
Brightness By Add	(-0.35, 0.35)	0.333553	0.327892
Contrast By Multiply	(0.01, 0.1)	0.295984	0.397236
Noise With FV	(0.5, 1.0)	0.220813	0.357522
Salt and Pepper Noise	(0.005, 0.01)	0.303765	0.352944
Speckle Noise	(0.8)	0.243307	0.853659
Gaussian Noise	(0.1)	0.245031	0.337980
Gaussian Sharpening	(2.0)	0.254306	0.257343
Shift Intensity	(0.1, 0.4)	0.311274	0.347057
Smooth Field Contrast Adjust	(2, (0.5, 4.5))	0.315087	0.297795
Adjust Sharpness	(-0.35, 0.35)	0.269351	0.374967
Brightness and Contrast Jitter	(0.1), (0.1)	0.282096	0.343027
Gaussian Blur	(5, (0.1, 0.5))	0.269272	0.357830
Erasing	((0.001, 0.01), random)	0.254103	0.344821

Table 4.3: Results for pixel-level transforms

4.1.4 Summary of Single Data Augmentation Methods

The evaluation of various augmentation techniques reveals that they significantly enhance model performance compared to models trained without augmentation. Affine transformations, including reflection, rotation, and translation, consistently improved model performance, with rotation and translation showing the most substantial gains. Pixel-level transformations had mixed results; noise with fixed values generally underperformed, while salt and pepper noise, speckle noise, and Gaussian noise provided more stable improvements. Brightness, contrast, and sharpness adjustments also contributed positively, especially brightness adjustments. Grid distortion and elastic transforms proved to be highly effective, significantly boosting robustness and stability. Erasing techniques, particularly with random replacing values, showed potential for improvement. Overall, the results emphasize the importance of augmentation techniques in enhancing classification accuracy and model stability.

4.2 Evaluation of Combined Data Augmentation Techniques

After analyzing the effectiveness of individual data augmentation methods on our dataset in the first part of the experiments, the second part focuses on evaluating the combinations of the best-performing methods. We selected four series, each comprising five methods, based on their performance relative to folds using the F1 metric. These series are visualized in Table 4.4, which also includes their evaluation based on F-1 Mean and F-1 Std metrics.

To test the impact of the order of methods within a series, we evaluated four additional series that are mirrored versions of the initial four. These mirrored series are also visualized in Table 4.4. In total, we tested eight combinations. The results indicate that the smallest F-1 mean value is 0.281269, which, although relatively low compared to other combinations, is still higher than the best run with any single augmentation method. This finding suggests that using even a suboptimal combination of methods is better than no augmentation, in terms of the F-1 mean metric.

When considering single augmentation methods, only Horizontal Reflection consistently improved performance across all parameter choices. Comparing single augmentation methods, some did result in higher F-1 mean values than the worst-performing combination. However, the best result obtained from a combination is 0.486309, which is more than 0.12 higher than the best result from any single data augmentation method, specifically rotation by 30 degrees in each direction. This combination nearly doubled the results compared to the best baseline configuration.

In the sub-experiment examining the effect of method order within a series, all mirrored order combinations outperformed their straight order counterparts. This indicates that the order of methods, in addition to the selected parameter or range of parameters, significantly influences the outcome. The mirrored combinations yielded F-1 mean values ranging from 0.370781 to 0.486309.

Methods	Parameters	Test F1 Mean		Test F1 Std	
		Straight	Mirror	Straight	Mirror
Translate Brightness and Contrast Jitter Reflection Contrast By Multiply Brightness By Add	(0.3, 0.3) (0.4), (0.4) horizontal (0.01, 0.1) (-0.35, 0.35)	0.381532	0.486309	0.295552	0.352499
Translate Elastic Transform Reflection Adjust Sharpness Contrast By Multiply	(0.3, 0.3) (5.0, 50.0) horizontal (1.5) (0.01, 0.1)	0.368172	0.439932	0.183921	0.323585
Shear Gaussian Noise Reflection Gaussian Blur Smooth Field Contrast Adjust	(-15, 15, -15, 15) (0.05) horizontal (5, (0.5, 1.0)) (2, (0.5, 4.5))	0.307746	0.370781	0.287813	0.360965
Brightness By Add Translate Shear Gaussian Blur Smooth Field Contrast Adjust	(0.005, 0.015) (0.3, 0.3) (-30, 30, -30, 30) (5, (0.5, 1.0)) (2, (0.5, 4.5))	0.281269	0.446705	0.196080	0.300755

Table 4.4: Results of different augmentation techniques on model performance, including reversed order (Mirror) experiments.

5 Conclusion

Medical imaging, particularly ultrasound, plays a crucial role in diagnosing and monitoring diseases. However, the limited availability of annotated medical image data poses a significant challenge in training effective deep learning models. Data augmentation offers a potential solution by artificially expanding the dataset, thereby enhancing the robustness and generalization of these models. This thesis addresses the need for improved augmentation methods tailored specifically for medical ultrasound data, a relatively under-explored area, especially for lung ultrasound images.

Currently, there is a significant emphasis on exploring the impact of data augmentation in medical imaging. Despite this, there is a relative scarcity of studies focusing specifically on ultrasound imaging, especially lung ultrasound. This is largely due to the novelty of the application and the absence of precise definitions and standardization in the experimental methods used. This gap highlights the need for more targeted research to develop and refine augmentation techniques that cater specifically to the unique challenges posed by ultrasound imaging.

This thesis implemented and evaluated various data augmentation techniques specifically for lung ultrasound images. The work aligns with existing literature by addressing the critical need for high-quality augmentation in medical imaging. The proposed methods were evaluated using a systematic approach, including parallel hyperparameter tuning and extensive validation through cross-validation techniques.

The evaluation of combined data augmentation methods reveals that using combinations of the best-performing augmentation techniques can further enhance model performance. The study found that even suboptimal combinations of methods performed better than the best single method. Additionally, the order in which these methods were applied significantly influenced the outcomes, with mirrored combinations consistently outperforming their original counterparts. These findings suggest that strategic combinations and orderings of augmentation techniques can lead to significant improvements in model robustness and diagnostic accuracy.

Future research directions suggested in this thesis involve exploring hybrid augmentation methods that combine multiple techniques to maximize their benefits and conducting extensive clinical validation to ensure the practical applicability of the proposed methods. Additionally, experimenting with other methods or variations of those experimented in this thesis could provide further insights. A more detailed study of the parameter spaces of these methods is warranted, as results have shown that they significantly influence the final outcomes. It is also worth exploring other strategies for applying the methods themselves.

Beyond the augmentation methods, improving results can also be achieved by selecting more suitable neural network architectures and fine-tuning their hyperparameters. Integrating domain-specific knowledge into augmentation processes to enhance their effectiveness is another promising area for further studies.

In conclusion, this thesis makes a significant contribution to the field of medical ultrasound imaging by enhancing the understanding and application of data augmentation techniques. The findings underscore the potential of these methods to improve diagnostic accuracy and model robustness, paving the way for further advancements in medical image analysis and machine learning.

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List of Appendixes